

5/8/2026

* Gauss - Markov Theorem: ($Y = X\beta + \varepsilon$, $\varepsilon \sim N_n(0, \sigma^2 I_n)$)
 If $a^T \beta$ is estimable, then $a^T \hat{\beta}$ is its unique BLUE,
 where $\hat{\beta}$ is any solⁿ to the normal equation:

$$X^T X \beta = X^T Y \quad \text{-- (i)}$$

Proof: $a^T \beta$ is estimable means, $\exists$ an LUE $l^T Y$ for $a^T \beta$,
 which implies that $a^T = l^T X$ for some $l \in \mathbb{R}^n$.

$$\text{Then } a^T \hat{\beta} = l^T X \hat{\beta}$$

If $\hat{\beta}$ is a solⁿ of (i), then $\hat{\beta} = (X^T X)^- X^T Y + z$, $z \in N(X^T X)$

$$\Rightarrow X \hat{\beta} = X (X^T X)^- X^T Y + X z$$

$$\Rightarrow X \hat{\beta} = H Y + X z$$

$$X z = 0 \text{ always (Why?)}$$

$$\therefore a^T \hat{\beta} = l^T H Y$$

$$\Rightarrow E(a^T \hat{\beta}) = l^T H X \beta = l^T X \beta = a^T \beta$$

Another representation:

$$a \in R(X) = C(X^T) = C(X^T X)$$

$$\Rightarrow a = X^T X u \text{ for some } u \in \mathbb{R}^p$$

$$\Rightarrow a^T \hat{\beta} = ~~u^T X^T~~ u^T X^T X \hat{\beta} = u^T X^T Y$$

$$\Rightarrow E(a^T \hat{\beta}) = u^T X^T X \beta = a^T \beta$$

Why is $a^T \hat{\beta}$ the BLUE?

If $h^T Y$ is a LZF, then $E(h^T Y) \equiv 0$

$$\Rightarrow h^T X \beta = 0 \quad \forall \beta$$

$$\Rightarrow h^T X = 0 \Rightarrow X^T h = 0$$

Write $a = X^T X u$, then

$$\text{Cov}(a^T \hat{\beta}, h^T Y) = \text{Cov}(u^T X^T Y, h^T Y)$$

$$= a^T X^T \text{Var}(Y)h$$

$$= a^T X^T \sigma^2 I_n h$$

$$= \sigma^2 a^T X^T h$$

$$= 0 //$$

$\therefore a^T \hat{\beta}$ is THE BLUE by previous theorem. $\square$

Sp. $Y = X\beta + \varepsilon$ where $\varepsilon \sim N(0, \Sigma)$ ($\Sigma \rightarrow$ positive definite)
(Come up w/ an appropriate version of Gauss-Markov theorem in this context.) -- (ii)

Q) How to characterize a BLUE for a (linearly) estimable linear parametric function $a^T \beta$?

$\rightarrow$ Observation: Under (iii), LSE of β and MLE of β are not necessarily the same.

$\hat{\beta}$ (LSE of β), satisfies $X^T X \beta = X^T Y$

$$(\hat{\beta}^{ML}, \hat{\Sigma}^{ML}) = \underset{\beta, \Sigma}{\text{argmax}} - \frac{1}{2} (Y - X\beta)^T \Sigma^{-1} (Y - X\beta) - \frac{1}{2} \log |\Sigma|$$

$$\Rightarrow \hat{\beta}^{ML} = \underset{\beta}{\text{argmin}} (Y - X\beta)^T \Sigma^{-1} (Y - X\beta)$$

$$\Rightarrow X^T \Sigma^{-1} X \beta = X^T \Sigma^{-1} Y \quad \text{-- (iii)}$$

If X has full column rank, then $X^T X$ & $X^T \Sigma^{-1} X$ are both invertible and $\hat{\beta}^{LS} = (X^T X)^{-1} X^T Y$
 $\hat{\beta}^{ML} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} Y$

Note: both $\hat{\beta}^{LS}$ & $\hat{\beta}^{ML}$ are unb. est. of β .

$$\text{Var}(\hat{\beta}^{LS}) =$$

$$\begin{aligned}\text{Var}(a^T \hat{\beta}^{LS}) &= a^T \text{Var}(\hat{\beta}^{LS}) a \\ &= a^T (X^T X)^{-1} (X^T \Sigma X) (X^T X)^{-1} a\end{aligned}$$

$$\text{Var}(a^T \hat{\beta}^{ML}) = a^T (X^T \Sigma X)^{-1} a$$

Hint: Spectral decomposition (by Shounak da)

* Let U_1 & U_2 be linear subspaces in $\mathbb{R}^n$.

- $U_1 + U_2 = \{u_1 + u_2 \mid u_1 \in U_1, u_2 \in U_2\}$ (sum)
- $U_1 \oplus U_2 = \{u \mid u = u_1 + u_2, u_1 \in U_1, u_2 \in U_2, u_1 + u_2 \text{ is unique rep}^n \text{ of } u\}$ (direct sum)

$U_1 \oplus U_2$ is possible only when, $U_1 \cap U_2 = \{0\}$.

- A linear projection into U_2 along U_1 :

Let $u \in U_1 + U_2$, then $u = u_1 + u_2$ for unique $u_1 \in U_1, u_2 \in U_2$. Then a linear projection into U_2 along U_1 is defined as $p(u) = u_2$. ("along" $\Leftrightarrow$ "parallel to")

Every linear projection p can be expressed in form of a matrix: $p(u) = P u$.

Propⁿ:

TFAE

(i) P is a projection matrix

(ii) P is idempotent ($P^2 = P$)

(iii) $\mathcal{N}(P) = \mathcal{C}(I - P)$

(iv) $\mathcal{C}(P) + \mathcal{C}(I - P) = \mathbb{R}^n$

(v) $\mathcal{C}(P) + \mathcal{C}(I - P)$ is a direct sum of subspaces.

- Orthogonal projections: for any subspace S of $\mathbb{R}^n$ the projection of x onto S along $S^\perp = \{y \mid y \text{ is orthogonal to } S\}$ is called the orthogonal projection $p(x)$ of x

$$e(I-p) = s^+ q^+ e(p) = s$$

$$\therefore P^T(I-P)=0$$

$$\Rightarrow p^* p = p$$

(i) P is an orthogonal projector

(ii) $P = P^T P$

(iii) $P^T = P$ & $P^2 = P$

- Representation in form of a basis. Let $S = \mathcal{C}(X)$ where X is $n \times p$. Then the orthogonal projection on $\mathcal{C}(X)$ can be represented as $P = X(X^T X)^{-1} X^T$ ($(X^T X)^{-1}$ is a symmetric g-inverse of $X^T X$)

- Review of g -inverse:

Defⁿ: If $A_{m \times n}$ and $G_{n \times m}$, then G is a g -inverse of A iff $AGA = A$.

Claim: Every matrix A has a g -inverse.

Proof: Full rank factorization of $A = P Q^T$ ($r = \rho(A)$)
 $n \times r$ $r \times m$

Q^T has right inverse, P has left inverse. So, we can construct G s.t. $AGA = A$ //

Propⁿ: Let $A_{m \times n}$ & $G_{n \times m}$. TFAE

(i) G is a g -inverse of A

(ii) ~~for~~ for any $y \in \mathcal{C}(A)$, $x = Gy$ is a solⁿ of $Ax = y$

(iii) AG is idempotent & ~~$G(AG) = G(A)$~~ $P(AG) = P(A)$

(iv) G_A is " " ~~P(A)~~ $P(G_A) = P(A)$

(iii') AG is the projection onto $\mathcal{C}(A)$ along $\mathcal{N}(AG)$.

Propⁿ: If $A_{m \times n}$ & $B_{n \times p}$ are matrices s.t. AB is defined, then

- (i) (a) if $P(AB) = P(A)$, then $B(AB)^{-}$ is a g -inverse of A .
- (b) if $P(AB) = P(A)$, then $(AB)^{-}A$ is a g -inverse of B .
- (ii) For any matrix A , $A^T(AA^T)^{-}$ & $(AA^T)^{-}A^T$ are g -inverses of A .
- (iii) If $\mathcal{R}(A) \subset \mathcal{R}(B)$ and $\mathcal{C}(C) \subset \mathcal{C}(B)$, then $AB^{-}C$ is invariant to the choice of B^{-} .

Propⁿ: Let $A_{m \times n}$ and $y \in \mathcal{C}(A)$. Then the class of solution of $Ax = y$ is given by $A^{-}y + (I - A^{-}A)z$ for any $z \in \mathbb{R}^n$.

Defⁿ: A^{-} is called a reflexive g -inverse if $A^{-}AA^{-} = A^{-}$.

Propⁿ: Sp. A^{-} satisfies $(A^{-}A)^T = A^{-}A$, then for any $y \in \mathcal{C}(A)$, $A^{-}y$ is a solution of $Ax = y$ w/ the min. norm. We then refer to such an A^{-} as a minimum norm g -inverse.

Propⁿ: Sp. A^{-} satisfies $(AA^{-})^T = AA^{-}$. Then for any x, y , $\|AA^{-}y - y\|_2 \leq \|Ax - y\|_2$, then A^{-} is called a least squares g -inverse.

Note: The Moore-Penrose inverse follows all 3 of these properties.